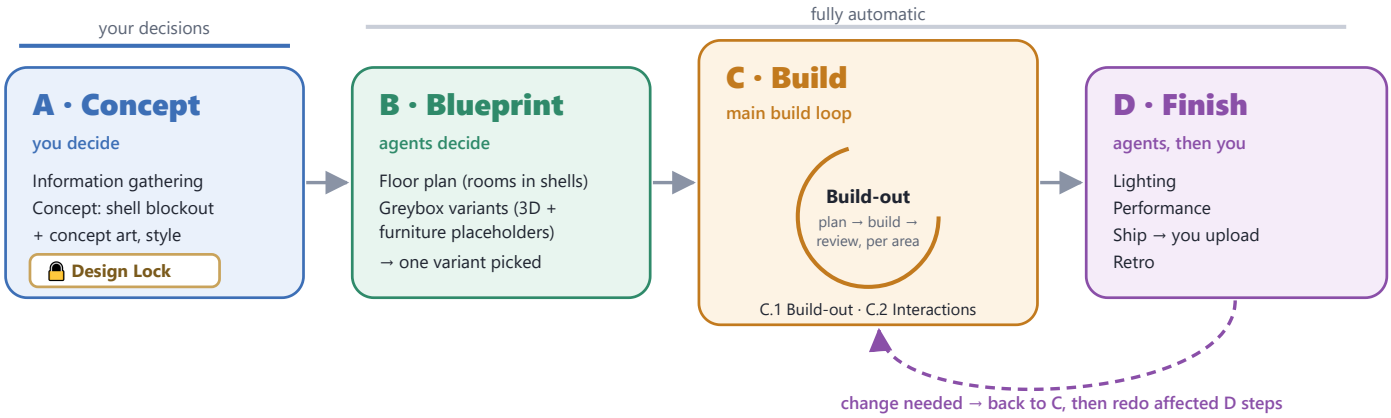
World Pipeline — Overview

Agentic VRChat World Development Kit (AWDK) · how a VRChat world goes from your idea to upload

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The four phases

You decide the style up front. After the Design Lock the agents build the whole world automatically; you sign off the upload at the end.



What happens in each phase

A Concept

You decide · ends with the Design Lock

1. **A.1 Setup** — project, Unity + MCP, packages, `WORLD.md` .
2. **A.2 Information gathering** — the model works from whatever you give it (description, design doc, reference pictures). **No fixed interview**: it only asks about what's missing — world kind (living space / event world) and event function, purpose, how many people, avatar eye height (default 1.2 m), cuddling → near clip, space types, mood, and **style**: palette, materials, lighting & time of day, era/place, exterior, furnishing feel. Pictures are saved and tagged.
3. **A.3 Concept: style track + layout track** — two separate tracks, you pick on each. **Style track (text → image only)**: text-to-image concept art from the brief and your tagged references, showing the lived-in, furnished world (never empty spaces); refined by changing the prompt text, no reviewer. **Layout track**: world-architect blocks out 2–3 shell layouts in Unity (massing only: buildings, trees, multi-level houses, bridges, paths, ground/water, sky; no interiors or furniture) and shows an eye-height arrival view, an overview and one intimate view as plain captures, no paint-overs; layout-reviewer ranks them and your pick becomes the geometry. Then text → image style rounds (palette, materials, lighting, conditioned on the lock images) until you approve. Images bind mood and style only.
4. **A.4** 🔒 **Design Lock** — brief + lock images (style track) + picked shell layout captures (layout track), approved once. Your last decision before the upload.

B Blueprint

Agents decide · optional review by you (on in trials)

1. **B.1 Floor plan (2D)** — inside the locked shells: which room is which, purposes, space types, rough sizes, what connects to what, public → private, doors, circulation; variants on different plan organisations, ranked by layout-reviewer. No furniture.
2. **B.2 Greybox (3D)** — 3–4 variant blockouts: building volumes, walls, heights, roofs, level changes **plus low-poly boxes for the big furniture** (couches, beds, tables, wardrobes, bar, screen, big rugs), true to size. No small props or wall decor. Reviewed and scored in 3D from eye height; **one variant is picked** and refined.
3. **Not playable yet** — no colliders, spawn, stair ramps, teleport doors or clip planes: those need the real objects and are built in C.2.
4. Furniture positions are exported **from the 3D scene** into the plan file and prop list.

C Build

Agents · the main build loop

1. **C.1 Build-out**, area by area, each pass doing what the area still needs:
   - **shell finish** — real walls, floors, roofs, windows, doors, trims, materials replace the greybox
   - **furniture swap** — each placeholder → the real asset at the same spot (moved only for collisions / better fit, reported)
   - **dressing** — secondary pieces and decor, by rule

   Assets are found or modelled as each pass needs them.
2. **C.2 Interactions & playability** — once the real objects exist:
   - **colliders** — walls, floors, big objects you mustn't walk through (trees, solid masses)
   - **lie-down colliders** on beds and couches, **off by default**: one world toggle lets non-full-body players lie on them; full-body players leave it off since they set their own height. Other props: no colliders
   - stair ramps, spawn, walk test
   - **Udon** — clickable teleport doors on the real doors, teleport islands, toggles (incl. the lie-down collider toggle), video, no world mirrors (personal mirrors; only real ones in bathrooms/gyms/dance rooms), pickups, VIP gate
   - clip planes and fog
3. Done when every area has SHIP IT with nothing left on its plan.

D Finish

Agents, then your sign-off · loops back to C

1. **D.1 Lighting** — draft bake → mood check against the boards → final bake → reflection probes immediately.
2. **D.2 Performance** — measure the worst viewpoints, find the cause, reversible fixes (asks before destructive ones).
3. **D.3 Ship** — remove helpers, build & test → **you upload**.
4. **D.4 Retro** — confirmed lessons; lock gaps become new information-gathering questions.
5. **Any change needed → back to C.** After the fix: rebake the area, rebake probes, re-measure, re-run the final review.

Plan → Build → Review: how every build stage works



Plan — a short pass plan: which area, the goal, what changes, which assets, what the reviewer will check. Logged.

Build — do exactly that plan.

Review — the stage reviewer's verdict with a **score out of 10** and the reasons. Its findings plus whatever the area still lacks become the next pass's plan.

Ends on SHIP IT with nothing left on the plan. Three rounds with no improvement on the same finding → change approach, or report blocked with evidence. Nobody waives a finding.

Rules that shape the pipeline

Images are feel references, never blueprints

AI images don't translate into 3D: perspective and parallax lines don't line up, scale is guesswork, layouts can be impossible. Builders and reviewers match **mood, palette, materials, lighting character and density** — never positions, proportions or layout. Geometry is judged only against the approved greybox and the spatial-design guide.

Scores out of 10

Every review and every ranking of variants gives a score out of 10 per item, with the verdict and reasons; the dressing reviewer scores style and build quality separately. Scores make rankings comparable, show progress across passes, and in trials are compared with your own score.

Furniture lives in 3D

Big furniture is placed as placeholder boxes in the greybox and judged from eye height; its positions are exported from the scene. Small props and decor come later, in Build-out.

Lock gaps

A style question during B–D that the lock didn't answer: the dressing reviewer decides provisionally against the boards, it's logged and listed in the upload report, and it becomes a new information-gathering question at the retro.

World kinds

Living space (hangout, home): real-architecture space types; a VIP supporter teleport room is optional (off unless the brief asks). **Event world** (e.g. a talk event, a race, an RP auction): function first, anything goes, game design allowed, no VIP room unless asked.

How involved you are

Preset	You decide
Design lock, then autopilot (default)	information gathering, concept (layout + look), style rounds → 🔒 **lock** · final upload
Every step (trials & calibration)	the default + a checkpoint after every stage, including the Blueprint review
Custom	the default + any stops you name

Trial runs

Checkpoint after every stage

Every decision, failed attempt, reviewer verdict and pass is logged with its reasons and screenshots (`Trial/log.jsonl`), and turned into a readable HTML/PDF report.

Your verdicts

At each checkpoint you mark each reviewer call agree / disagree (optionally with your own score out of 10), and say what to change and what worked. Every disagreement becomes a fix to a reviewer, builder rule or information-gathering question.

Calibrated

≥ 95 % agreement with the reviewers and no lock gaps for 3 trials in a row — then the pipeline runs unattended.

Trial worlds

Defined by you.

Who works where

Agent	Role	Phases
orchestrator (main session)	gathers information (asks only what's missing), owns WORLD.md, dispatches everyone	A B C D
world-architect	text → image concept art and style rounds, shell-layout blockouts shown as plain captures, floor plan, greybox variants, world kind & optional VIP island	A B
world-builder	Build-out, interactions & colliders, lighting	C D
layout-reviewer	shell layouts, floor plan, greybox and furniture layout in 3D, world size limits	A B
dressing-reviewer	Build-out passes and interactions: build quality **and** match with the boards (separate style / quality scores), lock gaps — never lighting	C
final-reviewer	lighting (mood vs the boards), final render, the only upload gate	D
perf-optimizer	measure, find the cause, reversible fixes	D
asset-finder · **blender-modeler**	find existing assets · model what doesn't exist	C

Always on: ZFightScan after every placement batch · console clean after every script change · WorldAudit (colliders, seats, pickups) · reflection probes right after every bake.

Spec: PIPELINE-LAYOUT.md · applied to the agents and skills · PDF: AWDK-pipeline.pdf